



José Fernando García Pérez

Senior Geospatial Data Engineer & Data Analyst

20+ years in GIS engineering · 6+ years in data analytics & engineering

CONTACT

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- Remote · USD / EUR · No sponsorship required

TECHNICAL SKILLS

Data & BI

Power BI (DAX, Power Query/M), dimensional modeling (star & bridge), embedded Mapbox maps, Looker Studio

Programming

Python (ArcPy, Pandas), SQL, JavaScript / React / HTML

Databases

PostgreSQL / PostGIS, SQL Server, Oracle, Supabase

GIS & Spatial

ArcGIS Pro / Desktop, QGIS, ModelBuilder, ANLA/IGAC geodatabases, MAGNA-SIRGAS CTM12, Mapbox

Cloud & Data Eng.

Azure, Databricks, Microsoft Fabric (in progress)

AI & Automation

RAG systems, Power Automate, n8n (in progress), Cursor AI, GitHub

LANGUAGES

Spanish Native

English B1–B2

Upper-intermediate, in progress — OLSA Academy (American English)

PROFILE

Geospatial Data Engineer pairing two decades of GIS engineering with modern data analytics and engineering, delivered across 150+ projects in government, infrastructure, environmental management, and risk assessment. I design dimensional data models over multi-million-record databases, build enterprise Power BI dashboards with complex DAX and embedded Mapbox cartography, and automate large-scale geospatial workflows with Python/ArcPy — architecting real-time pipelines that connect Oracle and PostgreSQL sources to interactive analytical platforms. Currently expanding into Azure, Databricks, and Microsoft Fabric.

20+

YEARS IN GIS

150+

PROJECTS DELIVERED

6+

YEARS IN DATA

KEY ACHIEVEMENTS

- Built a cadastral analytics platform** integrating ~4M property records (2015–2024) into a ~25-table dimensional model, with comparative temporal analysis and Mapbox geospatial visualization for the Municipality of Medellín.
- Architected a real-time environmental permit tracking platform** for the Aburrá Valley Metropolitan Area (AMVA), connected to an Oracle source, with automated cross-department response-time calculations across thousands of permit procedures.
- Engineered automated cartographic production workflows** in Python (ArcPy) and ModelBuilder — 1,000+ ANLA-compliant maps in under a year, cutting production time by ~85% versus manual work.
- Built a real-time urban tree inventory dashboard** for Medellín's Environmental Secretariat, linking a 1.3M+ record PostgreSQL database to interactive Mapbox cartography.
- Designed an automated PQRS case-tracking system** integrating SharePoint/Excel with Power BI in real time, with Power Automate delivering daily 9:00 AM overdue-case reports.
- Developed an AI-powered RAG assistant** enabling territorial-planning staff to query Medellín's Land-Use Plan (POT) in natural language.

EXPERIENCE

Independent Geospatial Data & Analytics Consultant

MAY 2011 – PRESENT

Self-employed — Medellín, Colombia

Independent consultant delivering GIS, data analytics, and data engineering services to government and infrastructure clients — 200+ projects, managing up to

EDUCATION

Postgrad. Spec. — Big Data & Business Intelligence

Universidad EIA · 2021

Postgrad. Spec. — Soil Mechanics

Universidad EAFIT · 2016

Postgrad. Spec. — Geographic Information Systems

Universidad de San Buenaventura · 2010

Civil Engineer

Universidad Nacional de Colombia · 2006

- 6 concurrent engagements. Contracting both directly and through my company, INGEOMATIC SIG SAS (incorporated 2014).
- ▶ **AMVA – Aburrá Valley Metropolitan Area** · GIS Specialist & Data Analyst · Apr 2024 – Present. Real-time environmental permit dashboards connected to Oracle; Python/ArcPy automation; MAGNA-SIRGAS CTM12 spatial processing.
 - ▶ **Municipality of Medellín – Planning Dept. (DAP)** · GIS Specialist & Data Analyst · May 2024 – Present. Power BI dashboards with dimensional modeling and embedded Mapbox; multi-criteria spatial analysis; AI-powered RAG assistant for the POT.
 - ▶ **DEVIMED S.A.** · GIS Engineer & Data Analyst · 2018 – 2026 · via INGEOMATIC SIG SAS. ANLA/IGAC geodatabases and Power BI for ANI/ANLA concessions; 1,000+ automated maps; Python predictive models across 40+ infrastructure projects.
 - ▶ **Bioexplora S.A.S.** · GIS Engineer, Data Analyst & Geotechnician · 2012 – 2024 · via INGEOMATIC SIG SAS. Power BI lead; 100+ GIS projects in risk management, land-use planning and environmental studies; 100+ ANLA/IGAC geodatabases.
 - ▶ **Government of Antioquia** · GIS Engineer & Database Specialist · 2012 – 2020. Departmental geodatabases; multi-criteria spatial models; ArcPy/ModelBuilder automation.
 - ▶ **IATECH / Medellín City-Center Mgmt.** · Data Analyst & GIS Engineer · 2021 – 2023. Financial dashboards for shopping centers; Python data pipelines; ArcGIS spatial analysis.

GIS Specialist & State Contracting Lead

FEB 2006 – MAY 2011

ARM Consultoría — Medellín, Colombia

- ▶ Led 30+ land-valuation studies, coordinating multidisciplinary teams and field crews.
- ▶ Served as technical GIS specialist and state-contracting specialist.
- ▶ Built spatial databases in PostgreSQL; migrated CAD datasets from AutoCAD to ArcGIS for government clients.

PROFESSIONAL DEVELOPMENT

Azure, Databricks & Fabric (2026) · Python Programming (2020) · Spatial SQL with PostgreSQL (2018) · SQL with MySQL (2018) · Python & ArcPy for ArcGIS (2018) · Project Management Diploma (2009)